

Roxana Fiorella Guillen Hurtado

guillen.fiorella@outlook.com | London, UK | +44 7448054155 | Portfolio: fiorellaguillen.github.io/portfolio/

Architect and geospatial data scientist with three years of experience in urbanism and sustainable mobility. Skilled in spatial analysis, quantitative research, and digital communication, applying data-driven approaches to support sustainable cities, climate action, and inclusive urban development.

EDUCATION

MSc Urban Spatial Science

Class of 2025 | UCL - The Bartlett Centre for Advanced Spatial Analysis

Master's programme focused on integrating data science and spatial analysis. Modules included Data Science for Spatial Systems, GIS, Quantitative Methods, Urban Simulation, Agent-based Modelling, Building Spatial Applications with Big Data, Remote Sensing, among others.

Bachelor's in Architecture

Class of 2021 | PUCP - Faculty of Architecture and Urbanism

11-semester programme with a strong focus on urbanism. Included modules in sustainable design, spatial analysis and urban planning. Awarded for an outstanding thesis on urban design, applying innovative approaches to vulnerable neighbourhoods.

Study-Abroad Semester | MSc. ITECH

2018-19 | University of Stuttgart

Semester in the Integrative Technologies and Architectural Design Research programme, focused on computational design, robotic fabrication, and architecture. Gained experience in parametric modelling, algorithmic design, and digital fabrication.

SKILLS

Technical skills

- **Programming & Data Science:** Python (Pandas, Geopandas, NetworkX, OpenCV), SQL, Git
- **Machine Learning:** Semantic Segmentation and Object Detection (YOLO, DeepLabV3+), Image annotation (LabelMe), Training of CV models, Scikit-learn, Pytorch, Applied ML for geospatial and urban data, Text mining, NLP.
- **Geospatial Analysis:** R, QGIS, Google Earth Engine, Network analysis, Spatial Interaction Models, Agent Based Modelling (ABM)
- **Data Visualization:** Ggplot2, Matplotlib, Leaflet, Bokeh
- **Architectural Software:** AutoCAD, ArchiCAD, Rhinoceros, Grasshopper (parametric design)
- **Design & Productivity:** Adobe Creative Suite, Microsoft Office and Google Workspace

Research skills

- **Research assistant** on climate adaptation and resettlement strategies in flood-prone Amazonian communities.
- Academic training in **quantitative methods** to analyse socio-demographic, mobility and spatial trends.
- Work experience on **qualitative urban analysis** to inform public sector urban policies.

Communication

- **Strong communication & public speaking skills**, actively collaborating as a Student Ambassador at The Bartlett
- **Delivered presentations** of my bachelor thesis research at academic events in Peru and Spain.
- **Conducted fieldwork and stakeholder consultations** with local authorities and communities.
- **Designed and facilitated workshops** for participatory urban development initiatives.
- **Curated and presented exhibitions** showcasing urban and policy proposals.

Languages

- **Spanish:** Native
- **English:** C1 level
- **German:** A2 level

AWARDS

2024 Bicentennial Generation Scholarship – Peruvian Government

Awarded to outstanding Peruvian professionals of high academic performance to pursue postgraduate studies.

International Grand Prize for Sustainable Urban Design - ADUS 2021

First prize awarded on the V Edition of the Architectural and Urban Sustainable Design Award for the Special Landscape Project for the Rimac River.

2022 & 2018 Excellence award - Faculty of Architecture and Urbanism PUCP

Received the Excellence award of the Architecture and Urbanism Faculty, for outstanding students of the faculty.

RELEVANT EXPERIENCE

03/2025 – 09/2025 | Dissertation Researcher

UCL-CASA & ERAco

- Conducting master's dissertation in a partner project between CASA and ERA Co., leveraging computer vision and crowdsourced street-level imagery to assess urban form in data-scarce cities in the Global South.

11/2024 – 09/2025 | Student Ambassador

UCL – The Bartlett Faculty of the Built Environment

- Represented The Bartlett in academic and outreach initiatives, engaging with prospective students and the university community.
- Created social media content and blog posts for The Bartlett's website, supporting public engagement and showcasing the MSc student experience.

06/2025 | Student Consultant – Building a Better Future Consultancy Project

Sutton Council & The Bartlett, UCL

- Collaborated with an interdisciplinary team to develop a regeneration proposal for Sutton Town Centre.
- Presented final proposal to Sutton Council representatives, gaining experience in public sector collaboration within the UK context.

07/2025 | Conference Assistant

CUPUM 2025

- Facilitated sessions at the 19th International Conference on Computational Urban Planning and Urban Management (CUPUM).
- Provided support to speakers and attendees throughout the event.

05/2025 | Participant – Data Storytelling Workshop

UCL Social Data Institute

- Participated in a workshop on data storytelling by UCL's Social Data Institute, focused on crafting clear narratives from complex datasets, developing skills to translate statistical insights into engaging, audience-specific visual and written stories.

09/2022 – 09/2024 | Visualization Coordinator

Municipality of Lima, Peru

- Managed a team of architects and urban designers to enhance the project's visual presence and public impact.
- Directed the visual communication strategy, tailoring materials for technical, institutional, and public audiences.
- Led the production of diverse visual content, including renders, videos, posters, monthly articles, and exhibitions.

10/2021 – 09/2022 | Urban Designer

Municipality of Lima, Peru

- Core team member in a multidisciplinary 170-hectare urban regeneration project for Lima's Historic Centre.
- Led public space design proposals (plazas, streets, riverfronts), integrating urban analysis, and mobility studies.
- Coordinated multi-stakeholder engagement with residents, authorities, and private actors to inform urban design proposals and planning documents.

06/2021 – 12/2021 | Junior Urban Designer

Cordillera Sur - Architecture Firm

- Contributed to public space and landscape projects focused on ecological and cultural sustainability.
- Designed visual communication materials for competitions and public presentations.

01/2018 – 08/2018 | Research Intern

HOME [Self-Sustainable Amazonian Cities] (CIAC-PUCP) - Research project

- Supported research on climate-induced resettlement and urban design strategies for the Peruvian Amazon.
- Contributed to the development of design guidelines and visual materials for two research publications and one policy guide.

12/2021 | Guest Speaker

Friedrich Ebert Foundation & 'Acuerdo por Iquitos' Foundation

- Presented two lectures in the forum 'A better urban future for Iquitos':
- Inhabiting the river: alternative urban model for the neighbourhood of Bajo Belen, Iquitos.
- Inhabiting the river: rethinking housing and the Amazonian neighbourhood in Bajo Belen, Iquitos.

10/2021 | Guest Speaker

UPC - Polytechnic University of Catalonia

- Presented a lecture in the module "Architecture and Cooperation 2021" given at the ETSAV-UPC.
- Lecture: Inhabiting the river: towards an alternative urban model for the flood-prone neighbourhood of Bajo Belen.